

nected to status LED, freewheeling diode, and inductive coil.

Optocoupler IC(4N25). The optocoupler's true function is to provide proper isolation between the DC power of relay and the input signal. It has an

internal LED across its pins 1 and 2, which is used to trigger base of the transistor to produce the same signal across output terminals 4 and 5 of 4N25. In this way, we get the same output signal but isolated from the input.

Switching transistor. The relay module comes with switching transistors. More the number of channels, more would be the number of switching transistors. These switching transistors can control the relays through Arduino or microcontrollers like ESP32. Here, a single-channel relay is used, so only one transistor is required. The switching transistor's function is to amplify the current to meet the requirement of the relay.

Status LED. There is an SMD type status LED on the relay module. Each relay has its own status LED, which turns on whenever the respective relay is energised through the input pin.

Freewheeling diode. There is a freewheeling diode to control the back emf produced by inductive coil of the relay. Back emf can cause severe damage to the relay driver circuit, so a freewheeling diode is used to counter the effect.

DC water pump. This pump is connected between NO and NC contacts of the relay. When the sensor detects hands, it gives a signal to the Arduino, which provides 5V signal to the relay driver circuit through digital pin D8 of Arduino. So, the relay is energised and pump switches on for ten seconds (as programmed) to discharge liquid from the hand sanitiser. After that the relay de-energises and the pump switches off. Potentiometer VR2 can be used to change the duration of pump remaining on.

Software

Arduino IDE 1.8.15 is used to write, compile, and upload the source code to Arduino board. The program code *Dispenser.ino* is written in Arduino programming language. The flowchart of the program is shown in Fig. 5.

Arduino Nano has pre-burned bootloader to upload the code to Arduino, so external hardware

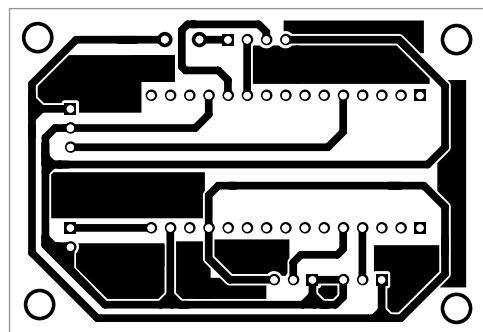


Fig. 6: Actual-size PCB layout

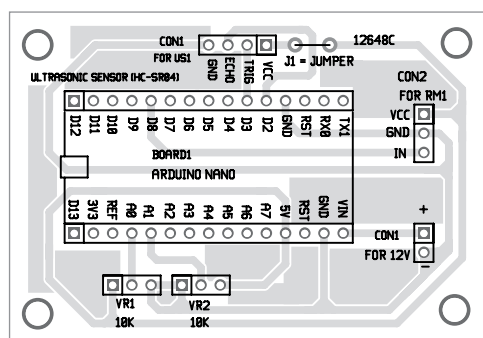


Fig. 7: Component layout of the PCB



Fig. 8: Proposed enclosure

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

(For the circuit on PCB)

Semiconductors:

Board1 - Arduino Nano board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

VR1, VR2 - 10-kilo-ohm potmeter

Miscellaneous:

- CON1 - 2-pin connector
- M1 - 12V DC water pump
- US1 - HC-SR04 ultrasonic sensor
- RM1 - 5V, single-changeover relay module
- Designed/general-purpose PCB
- Jumper wires

programmer is not required. A mini-A to mini-B USB cable is needed instead of a standard USB cable.

Compile the code and upload program to the Arduino Nano. Make sure you have chosen appropriate board and COM port from Tools menu for communication between Arduino and PC/laptop.

Construction and testing

An actual-size PCB layout for the circuit is shown in Fig. 6 and its component layout in Fig. 7. After successfully assembling the circuit on PCB and uploading source code in Arduino Nano, connect 12V DC power supply to the Arduino and pump. Then provide 5V DC power supply to the relay driver section and HC-SR04 sensor from 5V pin of Arduino.

Enclose the circuit in a suitable box as shown in Fig. 8. Properly mount the ultrasonic sensor at the bottom of the enclosure with the sensor facing downward. Adjust the time and range using respective potentiometers and put your hands below the sensor within the 10cm range. The sanitiser should flow out of the nozzle for ten seconds, after which the pump would switch off, until the sensor detects the next set of hands. **EFY**

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